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Дипломска работа

Способноста на мали јазични модели за промена на стил на македонски автори со примена на учење со поттикнување

[*Ментор: ...*] [*Кандидат: ...*]

Скопје, 2026

Abstract

[*Аспракт — written last (problem, method, the negative result with three or four numbers).*]

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Chapter 1

Introduction

1.1 Motivation

A poet's style is the part of a poem that survives a change of subject: the words they reach for, how long their lines run, whether they rhyme and how, which images return. Rewriting a poem so that its content stays and its voice becomes another poet's is a hard test of whether a language model has learned that layer of language rather than its surface. This is the task of *text style transfer*, here in its most demanding form: the "style" is not a register such as formal or informal, but one particular author.

Most work on style transfer and on language models in general is done in English, where data is plentiful. Macedonian is a low-resource language: it has few annotated corpora, few tools, and it is a small share of what general-purpose models are trained on. Poetry makes this worse. A poet writes a few hundred poems in a lifetime, often far fewer, so even a complete collection of an author's work is a small dataset by the standards of machine learning. The corpus this thesis is built on holds 1,239 poems and song lyrics by 38 Macedonian authors; the best represented author has 182 poems, the next two 160 and 134, and half of the authors have seventeen or fewer.

Large models can be prompted to write in a given style, but they are expensive to run and are not under the user's control. Small models of a few billion parameters can be trained and run on a single GPU, which makes them the practical choice for a language community without large computational resources. Whether they can learn a task as fine-grained as an individual poet's voice, from so little data, is the question this thesis sets out to answer.

1.2 Research question

The thesis asks:

> *Can a small language model, given only a source poem and the name of a Macedonian author, rewrite the poem in that author's style, and does reinforcement learning improve the result?*

Two parts of the question need making precise. "In that author's style" is measured in two ways that do not depend on each other: automatically, by stylometric features and by an author classifier that the training never sees, and by human judgement. "Reinforcement learning" is Group Relative Policy Optimization (GRPO) [24], applied on top of supervised fine-tuning with a reward built from the same stylometric profile of each author.

1.3 Approach and main finding

The work reached its final form after several approaches that did not work. Replacing selected words of the source poem with words typical of the target author, inflected to fit, preserves the poem but changes little of its style. Asking small local models to fill such slots fails at Macedonian itself. What did work was a large model, Gemma 4 31B [6], prompted with a detailed scaffold: a summary of the source poem, a prose description of the target author's style, a vocabulary palette and example lines, and explicit structural targets. That model became the *teacher*. Its outputs, about 115 per target author for ten target authors, are the training data for two *students* of similar size, Qwen3-4B [33] and Gemma 4 E4B, which see only the source poem and the author's name. Each student is fine-tuned with LoRA [9] on the teacher's poems and then trained further with GRPO.

The main finding is negative. The students learn the task's form: they write Macedonian, they write poems rather than prose, they change the source rather than copy it. They do not learn the voices. An independent author classifier recognises the intended author in 16.7% of the fine-tuned Qwen student's poems on the held-out test pairs, against 66.3% for real, unseen poems by the same authors; after GRPO the figure falls to 12.4%, even though the reward it optimises improves. The teacher, with the full description of the author in front of it, reaches 53.6%. [*Gemma student numbers on test; human ratings.*] The students' identified outputs belong almost entirely to the three authors with the most poems in the corpus, while the teacher also imitates some authors with fewer than forty.

The thesis argues that the limit is the data rather than the method. The corpus is enough for the teacher to imitate an author when the evidence of that author's style is in every prompt; it is not enough for a student to learn that style and keep it in its weights from about 115 examples per author. A stronger student, a better-tuned reward and reinforcement learning do not get past that limit.

1.4 Contributions

- A cleaned and part-of-speech-tagged corpus of Macedonian poetry, with poem-level identifiers, documented tagging corrections and a fixed train/validation/test split by author.
- Per-author stylometric profiles and a style-transfer reward built from them: part-of-speech distributions, lexical diversity, line and stanza structure, rhyme, characteristic vocabulary and an author classifier, with gates against copying the source and against degenerate output.
- A teacher–student pipeline that distils a large, heavily prompted model into two small students, followed by GRPO, run with matched settings for both students so that their results can be compared.
- An evaluation that separates real gains from reward gaming: the classifier used as a reward is never used to judge the outcome, a second classifier trained on different features is.
- A human evaluation of the teacher, both students and real poems on the same test pairs, with the real poems as a blind control.

1.5 Structure of the thesis

Chapter 2 reviews text style transfer, stylometry, knowledge distillation and reinforcement learning for language models. Chapter 3 describes the corpus and its preparation, and Chapter 4 the stylometric profiles and classifiers the rest of the work relies on. Chapter 5 summarises the earlier approaches and why they were abandoned. Chapter 6 presents the teacher, the students and the reward; Chapter 7 the evaluation protocol. Chapter 8 reports the results, Chapter 9 discusses why the students fall short, and Chapter 10 concludes.

Chapter 2

Background

2.1 Text style transfer

Text style transfer rewrites a text so that one attribute, its style, changes while its content stays [10]. What counts as style varies from task to task: sentiment, formality, politeness, or the manner of a particular author. The last kind is the oldest example in the field: Xu et al. [32] paraphrased modern English into Shakespeare's English and back, using the parallel modern versions of his plays.

Supervised transfer. When parallel data exists, pairs of texts with the same content in two styles, the task is ordinary sequence-to-sequence learning. The best known dataset of this kind, GYAFC, has about 110,000 informal–formal sentence pairs written by crowd workers [21]. Parallel data for an individual author's style practically never exists: nobody writes the same poem twice in two poets' voices.

Unsupervised transfer. Most of the literature therefore works with non-parallel data, texts in each style that are not translations of each other. One line of work separates content from style in a latent representation and decodes the content with a different style [25]. Another edits the text directly: it deletes the words that mark the source style and fills the gaps with words typical of the target style [14]. A third treats transfer as paraphrasing: a paraphrase model first strips the style, and a model trained on one style's texts then rewrites the neutral paraphrase in that style [12]. Reinforcement learning was used here as well, with rewards for the target style and for preserving content [17].

Prompting large models. Large language models can perform style transfer with no training at all, given an instruction and a few examples [22]. Infilling methods combine the editing view with prompting: the words that carry style are marked or masked, and a language model chooses their replacements, so that the rest of the text, and with it the content, is left untouched [15]. The second of the earlier approaches in this thesis (Chapter 5) follows this design.

Poetry and author style. Poetry adds constraints that prose does not have: lines, stanzas, metre and rhyme. Tikhonov and Yamshchikov [29] generated poetry in the style of particular authors, in English and in Russian, and evaluated the results both automatically and with human readers. Their models generate new poems from an author's corpus; the task in this thesis is harder in one respect, because the generated poem must also keep the content of a given source poem written by someone else.

2.2 Stylometry and authorship attribution

Stylometry measures style with countable features of a text, and its central application is authorship attribution: deciding who wrote a text [27]. The classic study of the disputed *Federalist* papers showed that the frequencies of common function words, which an author uses without thinking about them, identify the author better than the content does [18]. Burrows' Delta [1], a distance over standardised frequencies of the most frequent words, became the standard baseline. Modern attribution combines many feature types (word and character n-grams, part-of-speech patterns, vocabulary richness, sentence and line length) with a supervised classifier [27]. Vocabulary richness is length dependent, so a measure that does not grow with the text is needed to compare short poems with long ones; the moving-average type–token ratio (MATTR) computes the ratio over a sliding window of fixed size [3].

This thesis uses stylometry in two roles. A profile of each author's features defines what the reward asks for, and an author classifier measures whether a generated poem reads as that author's. When the same classifier is both rewarded and used to measure success, a model can learn to satisfy the classifier without writing like the author, so the evaluation uses a second classifier that the training never sees (Chapter 4).

2.3 Knowledge distillation and low-rank adaptation

Knowledge distillation trains a small *student* model to imitate a larger *teacher* [8]. For text generation the simplest and most effective form trains the student on the teacher's outputs, as if they were the reference data (sequence-level distillation, 11). Large language models have made this a common way of building training data: a strong model writes examples of a task, and a smaller model is fine-tuned on them [31]. The students in this thesis are trained the same way, on poems written by the teacher.

Fine-tuning all the weights of a model with billions of parameters needs more memory than a single GPU has. Low-rank adaptation (LoRA, 9) freezes the original weights and learns, for chosen weight matrices, a correction in the form of a product of two small matrices. The trainable parameters are a fraction of a percent of the model, and after training the correction can be merged into the weights.

2.4 Reinforcement learning for language models

Supervised fine-tuning teaches a model to imitate examples. Reinforcement learning instead lets the model generate, scores each output with a *reward*, and changes the model so that high-reward outputs become more likely. For language models the reward was first learned from human preferences between pairs of outputs [2, 34]; this reinforcement learning from human feedback (RLHF) is how instruction-following assistants are trained [19]. The optimisation algorithm in that work is Proximal Policy Optimization (PPO, 23), which needs, besides the model being trained, a second network of similar size to estimate the value of each state.

Group Relative Policy Optimization (GRPO, 24) removes that second network. For each prompt it samples a group of outputs, scores them, and uses each output's reward relative to the mean and standard deviation of its group as the advantage. A penalty on the divergence from a reference model keeps the trained model close to where it started. GRPO was introduced for mathematical reasoning and later used to train reasoning models with rewards that can be

checked automatically [7]. Here the reward is computed, not learned from preferences: it is the stylometric profile of the target author together with a measure of content preservation.

Any reward is only an approximation of what is wanted, and a model that optimises it hard enough finds ways to score well without doing the task. This *reward hacking* [26] grows with the strength of the optimisation [5], and it is why the reward and the evaluation in this thesis use different measurements. Training uses the TRL library [30].

2.5 Macedonian NLP

Macedonian is a South Slavic language written in Cyrillic, with rich verb morphology, a postposed definite article with three forms, and almost no noun cases. It has far fewer resources than English or than the larger Slavic languages. The CLASSLA project provides tagging and lemmatisation for South Slavic languages, built on the Stanza toolkit [20, 16, 28]; its Macedonian models assign each token a lemma, a universal part-of-speech tag, and a MULTEXT-East morphosyntactic description [4]. Large general-purpose models do write Macedonian, but with deviations in grammar and word agreement [35], and the small open models used as students in this thesis were trained on very little of it. The first open foundation model built specifically for Macedonian, with an 8-billion-parameter model, a corpus of 3.5 billion words, an instruction dataset and a seven-task benchmark, appeared only in 2025 [13].

Chapter 3

Data

3.1 The corpus

The corpus holds 1,239 Macedonian poems and song lyrics by 38 authors, 130,059 whitespace-separated tokens in all. Every poem has four fields: a numeric identifier, the author, the title and the text, in Cyrillic script with one verse per line and blank lines between stanzas. A typical poem is short: the median poem has 17 lines and 81 tokens.

The authors are very unevenly represented (Table 3.1). Three of them, Блаже Конески, Пијан Славеј and Ефтим Клетников, account for 476 poems, 38% of the corpus. The next author has 70. Half of the 38 authors have 17 poems or fewer, and eight have only one or two. Only 24 authors have at least five poems; these are the authors for whom a stylometric profile is built (Chapter 4), and the remaining 14 appear in the data only as poems, never as targets of style transfer.

Author	Corpus	Train split	Teacher examples
Блаже Конески	182	146	109
Пијан Славеј	160	128	108
Ефтим Клетников	134	108	110
Славко Јаневски	70	56	120
Влада Урошевиќ	61	49	114
Катица Ќулавкова	60	48	114
Братислав Ташковски	56	44	114
Веле Смилевски	49	39	118
Ристо Лазаров	48	38	120
Адем Гајтани	45	37	119
Other 28 authors	374	294	—

Table 3.1: Poems per author for the ten target authors: in the whole corpus, in the training split, and synthetic training examples written by the teacher model with that author as the target (Chapter 6).

3.2 Cleaning

Poem identifiers. An author and a title do not identify a poem: about fifteen (author, title) pairs are shared by several different poems, Блаже Конески alone having four poems titled *Песна*. Every poem is therefore given a stable numeric identifier, and every later stage, from the splits to the generated outputs, refers to poems by that identifier.

Script artefacts. 52 poems contain a Latin or Greek letter. In about 22 of them the letter is an artefact of digitisation: a character that looks like a Cyrillic one but has a different code point, such as a Greek *ο* standing for the vocative *о*, or words that mix two scripts, such as *што* or *КАМЕН*. The rest are intentional: Roman numerals and about thirty poems with whole words in Latin script. The two cases are kept apart. Artefacts are flagged during tagging and corrected by hand (below); intentional foreign words are left as they are.

3.3 Tagging and corrections

The corpus is tagged with CLASSLA-Stanza [*cite: Ljubešić and Dobrovoljc 2019; Terčon and Ljubešić 2023*], using its Macedonian models, which give for every token a lemma, a universal part-of-speech tag, a MULTEXT-East morphosyntactic tag [*cite: Erjavec 2012*] and morphological features. A hand check of 70 randomly chosen tokens against their lines put the accuracy of the part-of-speech tag at about 96%. The stylometric work uses only content words (nouns, proper nouns, verbs, adjectives and adverbs): 70,531 tokens.

Three corrections are applied on top of the tagger's output:

1. *Hand corrections.* A file of corrections, each keyed on a word and the tag judged wrong, is applied first, so that correctly tagged occurrences of the same word are never rewritten. A correction can change the tag, the lemma, or drop the token (for noise such as timestamps). There are currently 90.

2. *Relativisers.* The tagger labels the relativisers *што*, *кога*, *како*, *колку* and *каде* as adverbs. Left in, they were the five most frequent "content" words in the corpus, 1,478 tokens or 19% of all adverbs, and distorted every part-of-speech ratio built on adverbs. They carry a bare adverb tag without a degree, which real adverbs always have, and are filtered on that.

3. *Verbal adverbs.* Forms in *-јќи* (*барајќи*) are left unlemmatised by the tagger. They are re-lemmatised to their base verb; where the ending is ambiguous (*велејќи* from *вели*, but *знаејќи* from *знае*) the more frequent verb lemma in the corpus is chosen.

Every tagging run also writes an audit of suspicious tokens, 1,813 at present, grouped by reason: the two tag layers disagree on the word class (675 tokens), a lowercase word is tagged as a proper noun (430), the tagger gives no tag (357, mostly dialectal clippings such as *ко* and *сал*), a form is tagged differently from its majority reading elsewhere in the corpus (311), and script artefacts (40). The audit is the review queue for new hand corrections; nothing is filtered on it directly. Rules that flagged tokens for being rare were tried and dropped: in poetry rare words and lines without a verb are normal, and such rules were right about 5% of the time.

3.4 Splits

The poems are split into training, validation and test sets in the ratio 80/10/10 within each author, with a fixed random seed: 987, 126 and 126 poems. Splitting by author guarantees that every author with at least three poems has at least one poem in each set; the eight authors with one or two poems go entirely to the training set. The split is by poem, so two poems that share an author and a title can fall into different sets.

The split is computed once and read by every later stage. Everything that is learned from data (style profiles, classifiers, the teacher's examples, both students and the reward) uses training poems only. The validation set is used for model selection and for every comparison made during development; the test set is used once, for the final numbers in Chapter 8.

3.5 How little data there is

For a task as specific as writing like one poet, this is very little data. Even the best represented author has 146 poems in the training split, and seven of the ten target authors have between 37 and 56 (Table 3.1). By comparison, style-transfer benchmarks such as [*cite: GYAFC (Rao and Tetreault 2018)*] have tens of thousands of parallel sentence pairs for a single, much coarser style distinction.

The synthetic data does not remove this limit. The teacher writes about 115 examples for each target author, a balanced set, but each example is built from the same training poems: the description of the author's style, the vocabulary and the example lines in the teacher's prompt all come from them. An author with 37 poems in the training split is therefore represented by 37 poems' worth of style, however many examples are written from it. Chapter 9 returns to this: the students identifiably imitate only the three authors with more than a hundred training poems.

Chapter 4

Stylometric foundation

4.1 Author style profiles

[*to write*]

4.2 TF-IDF palettes

[*to write*]

4.3 The semantic knowledge graph

[*to write*]

4.4 Two author classifiers

[*The reward classifier and the independent judge.*]

4.5 Validation of the pipeline

[*Tagger migration, fixed bugs, community artifact.*]

Chapter 5

The path to a teacher

5.1 Word substitution

[*to write*]

5.2 LLM infilling

[*to write*]

5.3 A scaffolded prompt for a large model

[*Which becomes the teacher.*]

Chapter 6

Method

6.1 The teacher and the synthetic data

[*to write*]

6.2 Task format

[*to write*]

6.3 Supervised fine-tuning

[*to write*]

6.4 Reward

[*Style, content, gates, weights from within-group spread.*]

6.5 GRPO

[*Merge-then-LoRA, hyperparameters, v1 and v2.*]

6.6 The second student: Gemma 4 E4B

[*to write*]

Chapter 7

Evaluation setup

7.1 Protocol

[*to write*]

7.2 Metrics

[*to write*]

7.3 The judge

[*to write*]

7.4 A decoding baseline

[*to write*]

7.5 Human evaluation

[*to write*]

Chapter 8

Results

8.1 Earlier approaches against the students

[*to write*]

8.2 Qwen3-4B: validation and test

[*to write*]

8.3 Author identification per target

[*to write*]

8.4 The decoding baseline

[*to write*]

8.5 The teacher

[*to write*]

8.6 Gemma 4 E4B against Qwen3-4B

[*to write*]

8.7 Human ratings

[*to write*]

Chapter 9

Discussion

9.1 Why the students fail

[*Data scale, style collapse, nothing for the group advantage to amplify, reward gaming.*]

9.2 The teacher's own limits

[*to write*]

9.3 Threats to validity

[*to write*]

Chapter 10

Conclusion

10.1 Future work

[*to write*]

Appendix A

Examples

A.1 One example per stage

[*to write*]

Appendix B

Prompts and hyperparameters

B.1 Teacher prompt

[*to write*]

B.2 Student prompt

[*to write*]

B.3 Training configuration

[*to write*]

Appendix C

Reproducibility

C.1 Commands

[*to write*]

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